EL NIÑO ACTION BRIEFING · 15 OCTOBER 2026

West Jakarta, Jakarta

El Niño reaches West Jakarta as salt water in the Kali Angke intake now, then dengue in Cengkareng and Kalideres and floods on the Angke. First, make PAM Jaya warn before the next Hutan Kota cut and put tankers and lidded tanks in place before it comes.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Jakarta Raya (whole province, one row)	~61 for West Jakarta, scaled (DKI: 260)	6-month, Sep 2026 – Feb 2027	Pipe cuts now; dengue and floods from December

16/16 recommendations checked. 0 confirmed as written; all 8 actions corrected; 8 items of circulating advice contraindicated (the Do not cards: 5 province-wide, 1 of them precautionary; 2 West copies of province cards JK-X08 and JK-X09; and 1 West's own; the last 3 each corrected). The funding target fell 91% in verification, from the panel's Rp2.52 bn to Rp227 m: a flood standby with no unit cost or recipient was cut, and the drums were sized to TP PKK DKI's largest one-day handout on record (270 households). The set audit (15 October) cut the Rp225 m cadre top-up as a duplicate of what the province already pays. Exposure: Climate Impact Lab, *451,000 at Risk*, Sep 2026, pre-peer-review, heat only. West Jakarta's ~61 is our arithmetic, not an Impact Lab finding: 260 × 2,494,110 ÷ 10,669,702 = 60.8 (Dec–Feb: 147 × the same share = 34.4).

EXECUTIVE SUMMARY — AT A GLANCE



6.17°S 106.76°E, Indonesia · BPS boundaries via OCHA COD-AB; OpenStreetMap contributors (ODbL); Natural Earth

- **The risk.** Salt water stopped IPA Hutan Kota on 18 Sep, cutting 8 West kelurahan; Cengkareng and Kalideres hold 59% of 2026 dengue; 23 of the 40 deaths in BPBD's 2013 kelurahan table were in West.
- **First action.** PAM Jaya and JUP publish Hutan Kota's kelurahan and a 48-hour salinity warning by 31 October; BPBD stages tanks before cuts.
- **The money.** Rp88.5 m / Rp227 m / Rp280 m (US$4.9k / 12.6k / 15.6k); top placement: lidded drums in Cengkareng and Kalideres, 270 homes a round.
- **Do not.** Do not drill emergency deep wells in Kalideres and Cengkareng: the ground there sinks up to 7.9 cm a year.

What to do	**8** actions	p. 4
Who does what	RACI chart	p. 13
Where money does the most	funding	p. 13
Do not do this	**8** items	p. 15
What has happened here	the record	p. 18
How this was produced	methodology	p. 20
References	sources	p. 22

* Suggested formulation: check with a health worker before sharing.

West Jakarta (Kota Administrasi Jakarta Barat, Kemendagri 31.73) has 2,494,110 people (BPS, SP2020-based 2026 projection), eight kecamatan, and 56 kelurahan. Its poverty rate is 3.91% (BPS, March 2025). Tambora, at about 50,000 people per km² on the 2022 register (and over 100,000 in kelurahan Tambora itself), is among the densest places in Indonesia, and the register leaves out renters. The mayor is appointed and has no budget of its own: the province holds the money and PAM Jaya runs the pipes, so this briefing names the city offices (suku dinas) that do the work and leaves the city-wide items to the province file (`jakarta-province-*`). It is written as a proposed draft (*draf usulan*) for the two-year review of the RPB 2023–2027 (Pergub DKI 1/2024).

The season. An El Niño dry season that is already here: it has dried shallow wells and the ends of the pipe network and held smog over the city; it is expected to delay and shorten the rains; and it does not make the January–February floods less likely. Each briefing therefore covers one season, from dry wells, smog, and heat now, through dengue bred in stored water and the floods, to March 2027.

WATER AND AIR STATUS	READING	DATE
Jatiluhur reservoir (PJT II), most of Jakarta's raw water	90.73 m, about 16 m below its normal top; the level at which raw water to Jakarta is cut is **not published**	15 Oct 2026, 07:55 WIB
BMKG drought early warning, West Jakarta	**Siaga** (second level; Central, South, and North Jakarta and Seribu are at Awas; East not listed). Dasarian III not yet issued	Dasarian II, Oct 2026 (read 15 Oct)
BMKG drought early warning, September	BMKG put West, Central, East, and North Jakarta and Kepulauan Seribu at **Awas** for 11–20 September (BPBD DKI, via Kompas TV, 14 Sep); South Jakarta was not listed. BMKG's September bulletin is no longer online, and one report differs	Dasarian II, Sep 2026
IPA Hutan Kota (Kali Angke intake), the plant that failed	Seawater intrusion, TDS about 10,000 ppm against a 300 ppm limit; plant stopped; 8 West kelurahan cut, 72 tankers sent	18–21 Sep 2026 (latest found)
PAM Jaya outage reaching the east of West Jakarta	Pejompongan I substation outage: 17 West kelurahan cut, among them Tambora, Taman Sari, and Pal Merah	5–6 Jun 2026

* Suggested formulation: check with a health worker before sharing.

WATER AND AIR STATUS	READING	DATE
Tanker drops (reported, not complete)	72 PAM Jaya tankers to the Hutan Kota kelurahan	18 Sep 2026
Worst air reading found in the zone	ISPU 136, "tidak sehat", Cengkareng (GOR Cendrawasih) on udara.jakarta.go.id; the page's time stamp was inconsistent, so treat as **unverified**	Read 15 Oct 2026

Dengue by kecamatan (Dinkes DKI via Satu Data Jakarta): incidence in 2025 per 100,000, Cengkareng 196.4 (second of DKI's 44 kecamatan), Kembangan 133.3 (sixth), Kalideres 114.7, Kebon Jeruk 106.4, Taman Sari 72.0, Tambora 54.9, Grogol Petamburan 48.2, Pal Merah 43.0. In 2024 West had 3,698 suspected cases, the most of any city. In January–July 2026 it had 1,538 (Sudin Kesehatan, 5 August 2026): Cengkareng 548, Kalideres 361, Kebon Jeruk 160, Kembangan 111, Pal Merah 107, Grogol Petamburan 97, Tambora 90, Taman Sari 64, with no deaths reported. PAM Jaya publishes no service zones and its coverage figure (82.03%, May 2026) has no stated denominator, so this briefing does not estimate how many households are on wells.

ONE PLANT, ONE BELT

NAMED SECTION

The same handful of kelurahan in Cengkareng and Kalideres keep turning up. **Water:** IPA Hutan Kota, a 450–500 L/s plant in the Penjaringan urban forest (North Jakarta) run by PT Jakarta Utilitas Propertindo (JUP), takes raw water from Kali Angke. When the river's flow fell in September 2026, seawater reached the intake and PAM Jaya had the plant stopped: Cengkareng Barat, Cengkareng Timur, Rawa Buaya, Duri Kosambi, Kalideres, Pegadungan, Tegal Alur, and Kamal (and Kapuk Muara and Kamal Muara in North Jakarta), 44,872 customers. Katadata blamed the long dry season; PAM Jaya called it "not drought" but a supply fault. Either way it was the third Hutan Kota interruption in West in three years: repairs in September 2023 left Pegadungan fully cut, with yellow, salty wells, and maintenance cut seven kelurahan (about 55,272 customers) in July 2026. Rawa Buaya and Cengkareng also depend on bulk water from Tangerang; when it was cut in April 2026, PAM Jaya said "whoever lives at the end of the line gets what is left".

Floods: Duri Kosambi, Rawa Buaya, and Kembangan Utara flooded in all ten years in BPBD's record (2013–24); Cengkareng Barat and Tegal Alur in nine. **Ground:** Dinas SDA's GPS benchmarks put West's median at 4.15, 4.0, and 3.5 cm a year (2023, 2024, 2025), above the DKI median every year, with Cengkareng Barat at 7.9 cm in 2023, Jalan Kamal Raya (Kamal, Tegal Alur) at 6.7–7.5 cm in 2024, and Rawa Buaya and Kembangan Utara at 5–6 cm. None of Pergub 93/2021's groundwater-ban zones falls in West Jakarta. Neither the Semanan reservoir (begun April 2026, first cell due May 2027) nor the planned Duri Kosambi plant will change this season.

* Suggested formulation: check with a health worker before sharing.

WHAT THIS PLACE SHOULD DO RANKED

Ranked by expected lives saved per unit of effort before and through the peak. Backing lenses are shown per action. Convergence was checked for shared bias. Each action carries the RPB's columns: owner, place, target, date, and source of funds.

1. By 31 October, PAM Jaya and JUP publish the West kelurahan that IPA Hutan Kota feeds and agree a 48-hour salinity warning at the Kali Angke intake; BPBD Jakarta Barat names drop points with closed tanks in the eight kelurahan cut on 18 September

PARTIALLY CONFIRMED | EMERGENCY · HISTORIAN · SKEPTIC · PHILANTHROPIST | ADAPTATION INVENTORY: NOT SEARCHABLE THIS RUN | PHASE: PRA-BENCANA

OWNER	BY WHEN	COST	PREVENTS
PAM Jaya, with PT JUP; BPBD Jakarta Barat and Sudin SDA for drop points	List and threshold 31 Oct 2026; drop points before the next cut	Staff time; Rp70.1 m for 24 drop points with two closed tanks each (F02)	Unannounced cuts, and drums filled in a hurry and left open

PAM Jaya publishes no threshold for its raw water except the 300 ppm clean-water limit (Permenkes 2/2023), and in September TDS rose to about 10,000 ppm over one to two weeks before the plant was stopped. A warning set well below the cut level, at the intake, would have given the eight kelurahan days rather than hours. The tankers already exist (PAM Jaya sent 72 on 18 September; Dinas SDA holds 12 tankers, 5 mobile treatment units, and 22 kiosks for North and West, the first option to check for each drop point before any tank is bought); what is missing is the named place they go and somewhere closed to put the water. In 2023 BPBD trucked water to ten West kelurahan (Cengkareng Barat, Kapuk, Kedaung Kali Angke, Kalideres, Pegadungan, Tegal Alur, Kelapa Dua, Kembangan Utara, Palmerah, Jembatan Lima), more than any other city. Plan on at least 15 L per person per day (Sphere).

* Suggested formulation: check with a health worker before sharing.

Corrections: IPA Hutan Kota is in North Jakarta, not West; the 18 September cut hit 10 kelurahan, 8 in West, not 10 in West; the 2023 trucking article gives no cause, so the 2023 link to Hutan Kota rests on a separate detik report from Pegadungan. The 48-hour warning would be new: no PAM Jaya trigger exists. Province rule: JK-03 (posted tanker schedules in every city). Plants: `IPA-HUTANKOTA` (a new code; the plant-to-kelurahan feed is from the September notice), plus Tangerang bulk water for Rawa Buaya and Cengkareng.

2. From November, put a tight lid on water stored during the cuts in Cengkareng and Kalideres, with weekly cover-and-scrub rounds by jumantik and Dasawisma cadres; start with 270 homes

PARTIALLY CONFIRMED | PUBLIC HEALTH · PHILANTHROPIST · SKEPTIC | ADAPTATION INVENTORY: NOT SEARCHABLE THIS RUN | PHASE: PRA-BENCANA

OWNER	BY WHEN	COST	PREVENTS
Sudin Kesehatan Jakarta Barat and Puskesmas Cengkareng and Kalideres, with jumantik and Dasawisma cadres; BAZNAS (BAZIS) DKI buys the drums and TP PKK DKI delivers them	**270 homes with drums by 15 Dec 2026; rounds weekly to March 2027**	**Rp107 m at target for drums (F01)**	**The water stored for this season's cuts becoming the January–May dengue peak**

Cengkareng and Kalideres had 909 of West's 1,538 dengue cases in January–July 2026 (59%), and Cengkareng's 2025 incidence was the second-highest in DKI. These are also the kecamatan where the pipes were cut, so households filled every container they had. Last season West's monthly cases rose from 79 in October 2024 to 355 in May 2025. A lidded 160 L drum costs about Rp198,000 (US$11; the province's shared unit cost, before delivery). Drums go only to households not on PAM Jaya and TP PKK's free 300 L household-tank list (target 2,000 in 2026; in West, 100 went to Angke in January and 150 to Kamal in February), and every drum comes with the weekly line from province action JK-02: cover it, drain it, scrub the walls, where the eggs stick. The best trial evidence is Camino Verde (BMJ 2015; Nicaragua and Mexico): community mobilisation cut recent dengue infection in children by 29.5% (95% CI 3.8–55.3).

Corrections: Camino Verde mobilised households and did not hand out drums; the Girardot covers (Quintero 2015, US$19.2 a household) were measured on pupae, not dengue. Releases of Wolbachia mosquitoes in West are documented only in Kembangan; one unopened headline suggests caretakers in Cengkareng, so "no Wolbachia in Cengkareng" is unverified. BAZNAS DKI has no recorded drum distribution, so the drums are sized to TP PKK DKI's largest one-day handout on record: 270 households (with PAM Jaya, North Jakarta, 15

* Suggested formulation: check with a health worker before sharing.

Apr 2026), one round at the floor and up to three at the ceiling. TP PKK DKI is the delivery partner: with PAM Jaya it gave free 300 L tanks to 100 households in Angke in January and 150 in Kamal in February. Prices are undated retail listings: release money against a supplier quote.

3. By 30 November, every puskesmas kecamatan triages dengue warning signs, with written referral routes for Tambora, Grogol Petamburan, Pal Merah, and Kebon Jeruk, which have no RSUD

PARTIALLY CONFIRMED | PUBLIC HEALTH · EMERGENCY | ADAPTATION INVENTORY: NOT SEARCHABLE THIS RUN | PHASE: PRA-BENCANA

OWNER	BY WHEN	COST	PREVENTS
Sudin Kesehatan Jakarta Barat; the eight puskesmas kecamatan; RSUD Cengkareng, Kalideres, Kembangan, and Tamansari	**30 Nov 2026, with drills in December**	**Staff time and stock in existing budgets; not costed**	**Dengue cases becoming deaths in the January–May peak**

West can do little before March about how many people catch dengue; it can do a great deal about how many die. Indonesia's dengue case fatality rate fell from 0.9% in 2021 to 0.42% in 2025 (Kemenkes), which Kemenkes credits to clinical care and to source reduction by jumantik cadres; this action is the clinical half. Use the Kemenkes clinical guidelines (PNPK: KMK HK.01.07/MENKES/9845/2020 for adults, 4636/2021 for children), which follow WHO 2009: recognise the warning signs, monitor haematocrit, give fluids, refer early. West's four RSUD are in Cengkareng (Kelas B), Kalideres, Kembangan, and Taman Sari; Tambora, Grogol Petamburan, Pal Merah, and Kebon Jeruk have none, so write their routes now, to a named RSUD or a contracted private or national hospital. From January, after floods, the same alert covers leptospirosis: early treatment for fever after floodwater (province action JK-06). West had the most leptospirosis patients in DKI in 2014–16 (Dinkes; press report).*

* Suggested formulation: check with a health worker before sharing.

Corrections: the RSUD list is from 2022 and searches found no newer RSUD in the four kecamatan, which is absence of evidence; check Dinkes DKI's 2026 list. Leptospirosis prophylaxis is unproven (Cochrane), so the measure is early treatment.

4. By 30 November, name a shelter and send an evacuation and electrocution warning, tied to the Cengkareng Drain and Kali Angke siaga levels, for the kelurahan that flood nearly every year or lost the most people in 2013

PARTIALLY CONFIRMED | EMERGENCY · HISTORIAN · SKEPTIC | ADAPTATION INVENTORY: NOT SEARCHABLE THIS RUN | PHASE: PRA-BENCANA

OWNER	BY WHEN	COST	PREVENTS
BPBD Jakarta Barat and the wali kota; Sudin SDA for gauge levels; PMI Jakarta Barat; kelurahan and RT heads	**30 Nov 2026**	**Staff time**	**Drowning and electrocution in the January–February floods**

The kelurahan are Duri Kosambi, Rawa Buaya, and Kembangan Utara (flooded in all ten recorded years, 2013–24), Cengkareng Barat, Tegal Alur, and Duri Kepa (nine), and Kedaung Kali Angke, Tanjung Duren Utara, and Kota Bambu Utara (three deaths each in BPBD's 2013 kelurahan table). In January 2013, 23 of the 40 deaths in that table were in West Jakarta; BNPB counted 5 electrocution deaths among DKI's first 11 (18 January), 6 of 14 a day later, and one person was swept into Kali Sekretaris. The warning says: switch off at the meter before water enters, and stay out of water near cables. On 23 January 2026 the Cengkareng Drain reached 380 cm, above the 300 cm of Siaga 1 (the deputy public works minister, via Metro TV), and water stood 150 cm deep in Rawa Buaya. Tie the warning to Dinas SDA's siaga levels for the Cengkareng Drain and Angke Hulu gauges, so it goes out before the water arrives.*

* Suggested formulation: check with a health worker before sharing.

Corrections: "most of the 2013 dead were electrocuted" overstated the early count (BNPB: 5 of the first 11, 6 of 14 a day later), though by 21 January BNPB named electrocution the most common cause. Rawa Buaya did have posts in January 2026 (SDN 07 and a mosque), so the action names and stocks them rather than creating new ones. No official threshold table for the Cengkareng Drain was found; the only level found at source is 300 cm for Siaga 1 (Metro TV, 23 Jan 2026), so the siaga levels must come from Dinas SDA. West's January 2020 toll is unverified (BNPB snippets say 4).

5. Through the dry months, Sudin Gulkarmat Jakarta Barat checks wiring in the Tambora kampung that burned in 2024–26, finishes the extinguisher-per-RT programme there first, and maps water points other than hydrants

PARTIALLY CONFIRMED | EMERGENCY · FRONTLINE · SKEPTIC | ADAPTATION INVENTORY: NOT SEARCHABLE THIS RUN | PHASE: PRA-BENCANA

OWNER	BY WHEN	COST	PREVENTS
Sudin Gulkarmat Jakarta Barat, with RT heads and SKKL neighbourhood fire groups; PLN consulted	**Checks from late October; extinguishers in every RT of the 4 kelurahan by 31 Dec 2026**	**Existing Gulkarmat budget and the province extinguisher programme**	**Night fires in lanes no truck can enter, burning with no water in the mains**

West had 260 of DKI's 951 fires from January to 20 July 2025, the most of any city, and 635 of the 951 were electrical. A Tambora fire killed five people on 15 October 2024; Duri Utara lost 86 houses on 21 July 2025 (406 people displaced); Jembatan Besi burned at 04:19 on 11 January 2026; Krendang lost 27 houses on 28 May 2026. The province's one-extinguisher-per-RT programme ("1 RT 1 APAR", Ingub 5/2025) covers about 30,679 RTs, and the governor has said RTs in Tambora that had them saw smaller fires; coverage is not reported, so ask Gulkarmat for Tambora's. The dry season adds the water half: the 5–6 June Pejompongan outage cut Tambora, Taman Sari, and Pal Merah, so the map of waduk and kali the trucks can draw from matters when the hydrants are dry.

* Suggested formulation: check with a health worker before sharing.

Corrections: the cited source (Disway) gives the programme as one extinguisher per RT, not "at least two" as an earlier draft said. Fires here are mostly electrical and cooking fires all year; the El Niño link is the water supply and the dry fuel, not the ignition. Extinguisher prices were not verified, so nothing is priced.

6. From November, cadres check weekly on older people with hypertension or diabetes who live alone, in Tambora, Taman Sari, and Cengkareng, and daily on hot, smoggy, or water-cut days; each has two named neighbours who also move them in a flood

PARTIALLY CONFIRMED | PUBLIC HEALTH · FRONTLINE | ADAPTATION INVENTORY: NOT SEARCHABLE THIS RUN | PHASE: PRA-BENCANA

OWNER	BY WHEN	COST	PREVENTS
The puskesmas kecamatan, with posbindu and posyandu-lansia cadres, majelis taklim, and RT heads	Two neighbours named by 30 Nov 2026; checks to March 2027	Cadre time; no donor money (the Rp225 m top-up was cut in the set audit)	Heat, smog, and flood deaths among isolated chronic patients

Heat is the one hazard with a number: about 61 excess heat deaths for West Jakarta from September to February, by our scaling of Impact Lab's province row. The triggers are BMKG's daily maximum forecast at 34 °C or more, the nearest SPKU station at "tidak sehat" (province action JK-07), and any pipe cut, when an elder alone cannot carry water. DKI has Indonesia's highest diabetes rate (3.9% of people 15 and over, SKI 2023). The visit covers water, medicines, signs of heat illness, and from December the dengue warning signs.*

* Suggested formulation: check with a health worker before sharing.

Corrections: Semenza et al. (NEJM 1996, Chicago) is a case-control study of risk factors (living alone, odds ratio 2.3), not a trial of check-ins; treat this as a low-cost measure without a proven effect here. The Rp750,000 a month under Kepgub 497/2025 is an operational fund for Dasawisma group coordinators, from 1 September 2025, not pay to every cadre; jumantik receive about Rp500,000 a month (a 2015 provincial plan). The check-ins run on that existing support, so the cadre top-up was cut in the set audit.

7. By 30 November, RT heads in five Tambora kelurahan keep a count of renters, with no Jakarta ID card needed, and name a musala or mosque shelter in each dense RW

PARTIALLY CONFIRMED | FRONTLINE · FOOD SUPPLY | ADAPTATION INVENTORY: NOT SEARCHABLE THIS RUN | PHASE: PRA-BENCANA

OWNER	BY WHEN	COST	PREVENTS
Lurah and RT heads in Tambora, Duri Selatan, Duri Utara, Krendang, and Jembatan Besi, with mosque and musala boards; Sudin Sosial consulted	**30 Nov 2026**	**Rp50 m for five coordinators, once a recipient is named (F03)**	**Renters missing from every headcount after a fire, a flood, or a tanker drop**

After the Krendang fire of 28 May 2026, the camat said about 85% of survivors were renters or non-permanent residents. People sheltered in two musala and a tent; after the Duri Utara fire in 2025 they went to the RW hall and Masjid Al-Hidayah. KJP Plus and the KLJ card require a Jakarta ID (KTP DKI), so the people most hit by a fire are the least visible to the systems that pay out after one. A count held by the RT, of rooms and people rather than names, lets BPBD, Sudin Sosial, and the tankers plan for the people actually there; names stay in the kampung, because a list held by officials reads as an eviction list. The density figures (kelurahan Tambora about 100,957 per km², Duri Selatan 87,226) come from the register and leave renters out.

* Suggested formulation: check with a health worker before sharing.

Corrections: the 85% is the camat's figure from initial data (50 households), while other outlets counted 113–115 households; "all social aid needs KTP DKI" was narrowed to KJP Plus and KLJ. No yayasan with a delivery record in Tambora was found, so F03 waits for a named recipient.

8. By 31 March 2027, Sudin Kesehatan Jakarta Barat confirms whether Wolbachia has reached Cengkareng and requests egg supply for Cengkareng and Kalideres, which carry 59% of West's 2026 dengue; no effect is expected this season

PARTIALLY CONFIRMED | PUBLIC HEALTH · SKEPTIC | ADAPTATION INVENTORY: NOT SEARCHABLE THIS RUN | PHASE: PRA-BENCANA

OWNER	BY WHEN	COST	PREVENTS
Sudin Kesehatan Jakarta Barat, with Dinkes DKI; Kemenkes and the World Mosquito Program consulted	**Written request by 31 Mar 2027**	**Staff time; releases paid by APBN and partners**	**The 2027–28 season repeating this one in the two highest-burden kecamatan**

West Jakarta is one of five national Wolbachia pilot cities (national MoU, 30 July 2026). What is actually deployed is in Kembangan: releases began on 4 October 2024 at RW 07 Kembangan Utara and reached Meruya Utara, Kembangan Selatan, Meruya Selatan (811 buckets, May 2025), Joglo, and Srengseng. Kembangan's January–May dengue cases went from 377 (2024) to 227 (2025) to 89 (2026), and Sudin Kesehatan credits Wolbachia. The 2025 target of all eight kecamatan was missed for lack of eggs. The randomized trial in Yogyakarta found 77.1% protective efficacy (Utarini et al., NEJM 2021). Province action JK-12 asks for the 2027 rollout; this action puts West's two highest-burden kecamatan in that request.

Corrections: the 2025 bottleneck was egg supply, not caretakers. Kembangan's fall is a before/after comparison with no control area, and dengue cycles from year to year. "Srengseng 60% Wolbachia by October 2025" is a press figure, not verified.

WHAT THE PROVINCE DOES FOR WEST JAKARTA — WRITTEN ONCE

These items are city-wide, so they are written once in the province file (`jakarta-province-*`, code JK) and not priced here.

- **JK-01, the decree.** West was at Awas for 11–20 September (BPBD DKI, via Kompas TV) but is at Siaga, not Awas, in BMKG's October dasarian II warning, so the decree's Awas trigger is not met for West today; Central, South, and North Jakarta and Seribu are at Awas, and

* Suggested formulation: check with a health worker before sharing.

East is not listed. A Keputusan Gubernur declaring siaga darurat kekeringan lets BTT be shifted (by pergeseran) into water lines and opens a request for BNPB Dana Siap Pakai; only tanggap darurat opens BTT's direct release.

- **JK-02, the line on every notice:** "tutup wadah air, kuras dan sikat seminggu sekali". **JK-03**, posted tanker schedules and "boil before drinking" labels. **JK-04**, PAM Jaya and PJT II publish the Jatiluhur level at which Jakarta's raw water is cut.*
- **JK-05, trigger cash:** no money is promised until BPKD and the Inspektorat give a written opinion; then Rp300,000 once per household per declared status, at most twice a season, through Bank DKI.
- **JK-06 to JK-09:** the dengue and leptospirosis clinical alert, same-day smog and heat alerts, the PLN rule for pump power, and SPHP rice outlets. **JK-10 to JK-12**, asks to central government: rice aid for January–March 2027, Dana Siap Pakai, and the 2027 Wolbachia rollout.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Wali kota	PAM Jaya	PT JUP	BPBD Jakbar	Sudin SDA Jakbar	Sudin Kesehatan Jakbar	Puskesmas kec.	Cadres	RSUD	BAZNAS DKI	Kelurahan & RT	Community orgs, mosques & majelis taklim	PMI Jakbar	Sudin Gulkarmat Jakbar	Dinkes DKI
1. Hutan Kota warning and drop points	I	A	R	R	R						C				
2. Lids and cover-and-scrub rounds		C				A	R	R		R	C				
3. Dengue triage and referral						A	R		R						C
4. Flood shelters and electrocution warning	R			A	C						R		R		
5. Tambora wiring, extinguishers, water points		I									R	R		A	
6. Elder check-ins, two neighbours						C	A	R				R			
7. Renter count and mosque shelters				I							A	R			
8. Wolbachia request for 2027						A									R

What the chart shows: Sudin Kesehatan Jakarta Barat is accountable for three of the eight actions and the puskesmas for a fourth, so one meeting between them covers half the list. The kelurahan and RT heads appear in five rows; they are the delivery layer for water, fire, and renters alike. Only action 1 needs a party outside the city government, PT JUP, and only actions 2, 1, and 7 need outside money. Not shown as columns: PLN UID Jakarta Raya (consulted on 5), Sudin Sosial (consulted on 7), Kemenkes and the World Mosquito Program (consulted on 8).

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR

Rp88.5 m (~US$4,900): 270 drums (one TP PKK round) and 12 drop points

TARGET

Rp227.0 m (~US$12,600)

CEILING

Rp280.5 m (~US$15,600)

LARGEST UNCERTAINTY

Whether TP PKK DKI can run more than one 270-household drum round in West before the peak

Floor = Rp53.46 m (270 drums) + Rp35.04 m (12 drop points × two tanks) = Rp88.50 m. Target = Rp106.92 m (540 drums) + Rp70.08 m (24 points) + Rp50 m (5 renter-count coordinators) = Rp227.00 m. Ceiling = Rp160.38 m (810 drums) + Rp70.08 m (24 points) + Rp50 m (5 coordinators) = Rp280.46 m (US$15,626). The Rp225 m cadre top-up was cut in the set audit: the province already pays Dasawisma

* Suggested formulation: check with a health worker before sharing.

coordinators (Kepgub 497/2025) and jumantik. Rate: JISDOR Rp17,948 per US$ (15 October 2026). **All amounts are readiness money; none is trigger-released and none is committed.** The target fell 91% in verification, from the panel's Rp2.52 bn: a Rp250 m flood standby with no unit cost and no named recipient was cut (it duplicates BTT and Dana Siap Pakai), and the drums were sized to TP PKK DKI's largest one-day handout on record (270 households a round). Not counted: the province trigger-cash line JK-F01; water delivered free by PAM Jaya, BPBD, and Dinas SDA; and PAM Jaya and TP PKK's free 300 L household tanks. **Public money:** the 2027 APBD ceilings were fixed when the KUA-PPAS was signed on 13 August 2026, and the RAPBD 2027 (Rp82.89 tn) was tabled on 13 October. None of these actions is a new APBD 2027 line. Public money for them can come only by reallocation within existing suku dinas lines (Sudin Kesehatan's mosquito-source control for the drums), from the 2026 BTT on a declared status (by pergeseran under siaga darurat, for the drop-point tanks), from the APBD-P 2027, or from BNPB Dana Siap Pakai.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
BAZNAS (BAZIS) DKI, delivered with TP PKK DKI through Dasawisma and jumantik cadres	Lidded 160 L drums for households in Cengkareng and Kalideres not on PAM Jaya's free-tank list (action 2)	Rp53.5 m / 106.9 m / 160.4 m	Pre-positioned stock	270 / 540 / 810 households (one to three rounds of TP PKK DKI's largest one-day handout, 270 on 15 Apr 2026) × Rp198,000 (undated retail, before delivery). TP PKK DKI delivers: in West it gave free 300 L tanks to 100 households in Angke (Jan) and 150 in Kamal (Feb 2026). Excludes households holding a free tank. Camino Verde: −29.5% infection through mobilisation	Partially confirmed
BAZNAS (BAZIS) DKI, with RW committees	Two closed 500 L tanks at each drop point in the 8 Hutan Kota kelurahan (action 1)	Rp35.0 m / 70.1 m	Pre-positioned stock	24 points (an assumed 3 in each of 8 kelurahan) × 2 × Rp1.46 m; floor 12 points in the 4 kelurahan katadata named. Check Dinas SDA's 22 water kiosks (North and West) first	Partially confirmed
A community organisation in Tambora, through a registered yayasan (not yet named)	Five renter-count and shelter coordinators, Nov–Mar (action 7)	Rp50 m, target only	Direct grant	5 × Rp2 m a month × 5 months (assumption). No recipient with a Tambora record found	Unverified

* Suggested formulation: check with a health worker before sharing.

The funder's one thing: the cheapest water in West Jakarta is a warning. PAM Jaya already sends tankers free and already gives tanks to small piped homes; what nobody pays for is the 48 hours between salt reaching the intake and the taps going dry, and the lid on the drum filled in those hours. That is why the first action costs staff time and the first placement is a lid.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives, money, or the window here. Cards 1 to 5 are province-wide (written once as JK-X01 to JK-X05). Cards 6 and 8 keep West Jakarta's detail of province cards JK-X08 (deep wells) and JK-X09 (kampung clearance); card 7 is West's own.

Do not stand the response down when the rains return

CONTRAINDICATED

The first heavy rain is the handover, not the end. Water stored for the dry season becomes breeding stock within one to two weeks, and West's dengue climbed from 79 cases in October 2024 to 355 in May 2025. West's worst flood deaths came in a neutral year (2013).

Instead: keep the drop points, rounds, triage, and shelters running to March 2027; stand down on falling dengue cases, not on rainfall.

Do not make fogging the visible dengue response, and do not bulk-buy temephos without a bioassay

CONTRAINDICATED

Jakarta *Aedes aegypti* resist permethrin, the fogging class (Hamid et al., PLoS ONE 2017), and Kebon Jeruk in West Jakarta was one of the sites tested. Kemenkes asks for fogging only after a case investigation. In Camino Verde, households using temephos had higher odds of child infection.

* Suggested formulation: check with a health worker before sharing.

Instead: fog only on a case investigation; put the effort into lids and weekly scrubbing (action 2); choose any larvicide after a WHO bioassay.

Do not tell households to "store water" without "cover it, drain and scrub it weekly"

CONTRAINDICATED

Aedes eggs stick to container walls and survive draining. Storage advice alone multiplies breeding sites in exactly the kelurahan where the pipes were cut in September.

Instead: every notice and tanker drop carries the whole line (province action JK-02).

Do not present tanker, well, or refill-depot water as drinking water *

CONTRAINDICATED PRECAUTIONARY

No water-quality data on West Jakarta tankers was found, and drums filled from them sit open for days; the same caution covers well water (Pegadungan's ran yellow and salty in 2023) and refill-depot water. This card rests on precaution, not on a documented outbreak.

Instead: label it "air bersih, bukan air minum: rebus dulu"; drink boiled or gallon water.

Do not answer smog with road spraying, mist generators, cloud seeding, or odd-even rules, and do not spend drought water on misting

CONTRAINDICATED

The health minister said in August 2023 that road spraying only moves pollution. Each mist generator uses water that West's network ends do not have. Cloud seeding needs clouds that a dry El Niño season does not bring. Odd-even plate rules, the province card's fourth item (JK-X05), protect no one at risk on a smog day. Smog is worsened by a dry El Niño season, not caused by it.

* Suggested formulation: check with a health worker before sharing.

Instead: same-day alerts and one closed, filtered room for high-risk patients (province action JK-07).

Do not drill or permit emergency deep wells in Kalideres and Cengkareng when the pipes run dry

CONTRAINDICATED CORRECTED

Cengkareng Barat sank 7.9 cm in 2023 against a DKI average of 3.9 (Dinas SDA), and Jalan Kamal Raya 6.7–7.5 cm in 2024. Wells in Pegadungan already ran yellow and salty in 2023. Pergub 93/2021 has no groundwater-ban zone in West Jakarta, so nothing stops an emergency bore, and every one deepens the floods in the same kelurahan.

Instead: warned, staged tankers and closed tanks (action 1), and pipe connections at the network ends.

Do not plan Tambora's fire safety around hydrants and fire trucks alone

CONTRAINDICATED CORRECTED

The fatal Duri Utara fire of 8 July 2023 was in a lane only motorcycles can enter; trucks came but could not reach the door. When Pejompongan is down, as on 5–6 June 2026, the hydrants in Tambora and Taman Sari are dry too.

Instead: extinguishers at every RT and gang mouth, hoses run in from the road, and mapped waduk and kali to draw from (action 5).

Do not use the flood season to clear riverbank kampung on Kali Angke and Pesanggrahan

CONTRAINDICATED CORRECTED

Riverbank homes in Kedaung Kali Angke and Kapuk were demolished for the Kali Apuran works (22 December 2014, by LBH's record); in August 2025, 116 households sued the mayor and the province in PN Jakarta Barat for more than Rp119 bn, saying they were never compensated. Angke and Pesanggrahan normalisation has stalled since 2017 on land, not on clearance. A flood warning that reads as an eviction notice is a warning people ignore.

Instead: map no finer than the kelurahan, warn through the RT, and keep the electrocution message about the meter, not the house.

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Jan 2013	Jakarta flood (neutral): Kali Angke, Sekretaris, and Cengkareng Drain overflow	23 of the 40 deaths in BPBD's kelurahan table were in West; BNPB: 5 electrocution deaths among DKI's first 11 (18 Jan), 6 of 14 a day later	Normalisation begun, then stalled on land (2017)
Dec 2014	Kali Apuran demolitions, Kedaung Kali Angke and Kapuk	Households displaced; 116 sued in 2025	Case outcome unknown
Jan 2020	New Year flood (El Niño under ONI v6)	Unverified for West: none in BPBD's table; BNPB snippets say 4; two drownings named in press	Nothing specific
Jul 2023	House fire, Duri Utara, in a motorcycle-only lane	1 dead	Nothing found
Sep–Nov 2023	El Niño dry season: Hutan Kota repairs and dry wells in Kalideres; BPBD trucks water to 10 West kelurahan	No deaths recorded	No status declared
Apr 2024	Leak on the PJT II raw-water pipe to Pejompongan, from 1 April (El Niño ending; West's part of province event JK-E18)	37 West kelurahan without water	Repaired
Oct 2024	Fire in a dense Tambora kampung (neutral)	5 dead, about 30 houses	Extinguisher-per-RT programme (Ingub 5/2025)
Jul 2025	Fire, Duri Utara	86 houses; 406 people displaced; 9 injured	Nothing found

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Jan 2026	Kali Angke and Pesanggrahan overflow (neutral)	31 RT flooded, 150 cm in Rawa Buaya; one death in Grogol Petamburan in a press snippet (unverified)	Nothing specific
May 2026	Fire, Krendang (El Niño developing)	27 houses; about 200 people, mostly renters	Nothing found
Jun 2026	Pejompongan I substation outage (West's part of province event JK-E20)	17 West kelurahan cut	Restored
Jul and Sep 2026	IPA Hutan Kota: maintenance (7 kelurahan, about 55,272 customers), then seawater at the intake (8 West kelurahan, 72 tankers; West's part of province event JK-E21)	No deaths reported	10% bill discount; no warning rule

The recurrence nobody has noticed: the kelurahan that lose their water when IPA Hutan Kota fails in the dry season (Pegadungan, Tegal Alur, Cengkareng Barat, Rawa Buaya, Duri Kosambi, Kamal) are the same ones that flood almost every year and sink fastest, and their wells are salty. One plant, run by one operator on one river, is the dry-season failure point for West's flood belt, and the same households face both halves of this season. ENSO phases follow NOAA's ONI (ERSSTv6, five-season rule). City-wide events (1996, 2002, and 2007 floods; 1997–98 rice; 1998, 2004, and 2024 dengue; 2023 smog) are in the province file; West's share of the 2007 toll is not broken out by Bappenas. The 2024 PJT II leak, the June 2026 Pejompongan outage, and the September 2026 Hutan Kota stoppage crossed city lines, so they are province events (JK-E18, JK-E20, JK-E21) and this table gives West's part only.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Is the Hutan Kota cut a drought problem?** Katadata reported it as the long dry season; Bloomberg Technoz quoted falling river flow and seawater intrusion; PAM Jaya called it "not drought". The action does not depend on the answer: a salinity warning works either way.
- **Lids and drums are the best use of a donor's money** (Philanthropist) versus **no trial shows lids save lives** (Skeptic and the verifiers). Camino Verde measured community mobilisation; Girardot measured pupae. The drums are funded as part of the weekly rounds and as a pilot.

* Suggested formulation: check with a health worker before sharing.

- **Wolbachia worked in Kembangan** (Sudin Kesehatan: cases down about 60%) versus **a before/after drop in a cycling disease proves little** (verifiers). Either way it cannot help this season, so it sits at the bottom of the list as a request for 2027.

WHAT NOBODY COVERED — BLIND SPOT

Who is on a well, and who rents. No survey says how many West households depend on groundwater, PAM Jaya's coverage figure has no denominator, and the registers that count people leave out the renters who were 85% of one fire's survivors. Nor did any lens find IPA Hutan Kota's operating rules: the salinity at which JUP stops it, the contract between JUP and PAM Jaya, or how many weeks Kali Angke can stay low before the next cut. If the rains are delayed into December, as BMKG expects for most of Java, a fourth cut this season is likely, and action 1 becomes the city's main water operation rather than a patch.

HOW TO READ THIS — METHOD

The panel is author-built. Simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale. Here the only row is the whole province (Jakarta Raya, 10.7 million people), so West Jakarta's figure is a population share. The El Niño projection covers *heat only*; water cuts, flood, fire, smog, and dengue are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED — METHODOLOGY

- **Scope.** The location was taken as given: Kota Administrasi Jakarta Barat, DKI Jakarta (Kemendagri 31.73, BPS 3174; 8 kecamatan, 56 kelurahan; 2,494,110 people, BPS 2026 projection). Climate Impact Lab impact region: GADM IDN.7_1 "Jakarta Raya", the whole province. Its row (about 260 excess heat deaths, September 2026 – February 2027; 147 in December–February) was scaled by population share (23.4%) to about 61, our arithmetic. Kelurahan is the finest unit named; no RW is ranked; dengue is shown no finer than kecamatan.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts applied to temperature–mortality relationships, against a 1996–2025 baseline. Country and province figures cover September 2026 – February 2027. Heat only; never added to the Impact Map's long-run projections.

* Suggested formulation: check with a health worker before sharing.

- **Expert panel.** The eight lenses were run as five research agents, each covering one or two lenses (water and emergency management; public health; frontline and food supply; economist and philanthropist; skeptic and historian), from web sources, to a fixed brief that carried the set's verified facts (PLAN.md log), so that province-wide items were not researched twice.
- **Local evidence.** From the project GIS base (`projects/elnino-jakarta/gis/jakarta_base.gpkg`), built from Satu Data Jakarta and joined by name within the parent unit (Satu Data misfiles Cengkareng under North Jakarta): BPBD flood events by kelurahan (2013–2024); Dinkes dengue by kecamatan (suspected cases 2024, incidence 2025); Dinas SDA subsidence benchmarks (2023–2025), joined to kecamatan and kelurahan by location; puskesmas. BMKG's drought warning was read at bmkg.go.id on 15 October 2026.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias. Contradictions were kept. Actions were ranked by expected lives saved per unit of effort, each was given a named payer or marked unfunded, and the placements were reconciled against the ranked list. City-wide items were left to the province file.
- **Verification.** Three verifier agents checked all 16 recommendations, the funding envelope, and key events: water, flood, and fire; health; and money, renters, and evictions (with the financial checks). Their corrections are shown under each action.
- **Revised 2026-10-15 after the set audit** (fixes applied: the Rp225 m cadre top-up cut as a duplicate of the Dasawisma coordinator fund and jumantik pay; 2024 Satu Data dengue figures labelled suspected cases; the 2013 electrocution count corrected to BNPB's (5 of the first 11, 6 of 14), and one person, not several, swept into Kali Sekretaris; the dengue fatality line, the extinguisher programme (one per RT), and the Cengkareng Drain reading restated to their sources; the BAZIS capacity anchor removed and the drum tiers resized to TP PKK DKI's largest one-day handout (270 / 540 / 810 households), with TP PKK DKI as delivery partner; Dinas SDA kiosks named as the first option for drop points; the free 300 L tanks given their target and West rounds; the September BMKG Awas list added; the PJT II leak dated 1 April 2024; the three cross-city outages referenced to province events JK-E18, JK-E20, and JK-E21; Do-not cards matched to the province cards, with 6 and 8 referenced to JK-X08 and JK-X09; the Krendang link repaired; UK spelling).
- **Who does what.** The RACI chart is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not searchable this run (the page needs a browser filter and its dataset is behind a download form). This is unsearched, not "none exists".
- **Health wording.** Passages where a wording error could hurt someone (dengue danger signs and care, heat, flood safety, and drinking-water treatment) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources,

* Suggested formulation: check with a health worker before sharing.

not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.

- **Rules.** No cost is derived by dividing a published total, and no local figure is invented. One exchange rate for the set: JISDOR Rp17,948 per US$ (Bank Indonesia, 15 October 2026). Every changing reading carries its date. Indonesian text for every action, funding line, and event is in `west-jakarta-barat-id.json`, not yet reviewed by a native speaker.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources; no one in West Jakarta was interviewed. Several sources were read as search snippets only and are marked so in the corrections. Run-specific gaps: IPA Hutan Kota's stop rule and the kelurahan it feeds in normal times; West's January 2020 and January 2026 flood deaths; whether Wolbachia has reached Cengkareng; RSUD in the four kecamatan without one (2026 list); the ISPU reading; West jerrycan prices; West's share of social-card holders; the drum and tank prices, which are undated retail listings.

References · Water: Hutan Kota salinity, Sep 2026, 18 Sep cut and compensation, Jul 2026 maintenance, katadata, 23 Sep 2026, Hutan Kota plant, Kalideres 2023, 2023 trucking, Tangerang bulk water, Duri Kosambi plant, Jun 2026 outage, Apr 2024 leak, leak from 1 Apr, free 300 L tanks, Dinas SDA kit, PAM Jaya tankers, subsidence 2023, Pergub 93/2021, Sphere · Floods and fires: 2013 dead, 2013 BNPB, 18 Jan, 2013 BNPB, 19 Jan, Cengkareng Drain, Jan 2026, Jan 2026 flood, Jan 2026 shelters, West pumps, Angke Hulu siaga, Satu Data Jakarta (BPBD, Dinkes, Dinas SDA tables), fires and extinguishers, Tambora 2024, Duri Utara 2023, Duri Utara 2025, Krendang 2026, Tambora density, eviction lawsuit · Health: dengue Jan–Jul 2026, 2024–25 season, Kembangan Wolbachia, egg shortage, Meruya Selatan, Wolbachia MoU, Camino Verde, Quintero 2015, Hamid 2017, PNPK dengue, dengue CFR, RSUD list, Semenza 1996, ISPU, road spraying · Money: KUA-PPAS 2027, RAPBD 2027, Perban BNPB 4/2020, drum price, tank price, JISDOR · Season: BMKG drought warning, September Awas list (Kompas TV), PJT II Jatiluhur, NOAA ONI v6
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory
Generated 15 October 2026 for Kota Administrasi Jakarta Barat, DKI Jakarta, Indonesia. Indicative, not for land or permit decisions. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

* Suggested formulation: check with a health worker before sharing.